

Vijay Raghuwanshi

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Backend and Big Data Engineer with over four years of experience designing distributed systems and large-scale data platforms. Built national-scale analytics systems that process billions of events daily while keeping services highly available. Developed social media intelligence platforms for law enforcement. Focus areas include optimizing query performance on massive datasets, creating secure microservices, and improving data pipeline efficiency.

Experience

- ClearTrail Technologies Pvt Ltd Indore, India
 - Software Engineer Jul 2024 – Present
 - Designed GCP Pub/Sub messaging and ELT pipelines for the Target Anomaly analytics platform, processing 5 to 10 billion daily telemetry events to support high-throughput intelligence workflows.
 - Developed a pipeline scheduling framework running 3,000+ daily jobs, writing complex Spark SQL queries using CTEs and built-in functions to optimize data processing and improve query speed by 40%.
 - Created a database rule engine that parses custom queries, using a PostgreSQL star schema to support a multilayered declarative system for complex, dynamic UI form rendering, reducing rule deployment times by 90%.
 - Modeled database schemas using a Medallion architecture comprising silver and gold layers, organizing telemetry records to support pattern identification and multi-dimensional relation mapping.
 - Maintained high-availability database clusters and scale-out storage systems to ensure 99.99% system uptime, zero service downtime, and reliable database failover recovery capabilities.
 - Developed a backend gateway and web-based dashboard to manage interactive database cluster sessions, optimizing driver memory allocation and reducing driver process memory overhead by 45%.
 - Designed and integrated a unified monitoring stack to track component health in real time, implementing reporting and automated alerting that reduced pipeline failure detection times by 35%.
 - Containerized and deployed modular microservices on Kubernetes using GitLab CI/CD pipelines, ensuring consistent application rollouts, structured code reviews, and zero-downtime releases.
 - Associate Software Engineer May 2022 – Jun 2024
 - Developed modular, high-throughput web scraping microservices in Go and Gin, implementing automatic credential rotation and proxy management to achieve a 98% success rate across social platforms.
 - Streamlined telemetry data extraction and enrichment pipelines using asynchronous workers and proxy pools, scaling the architecture to handle over 50 million daily records with high reliability.
 - Engineered a secure Python microservice for file validation and sanitization, running check pipelines entirely in memory to prevent malicious uploads while maintaining processing latencies under 50ms.
 - Created high-fidelity synthetic log generators in Python modeling network traffic and carrier-grade address allocations, helping stress-test system capacity and reducing QA testing times by 40%.
 - Built autonomous web crawlers and data collection systems using Playwright, developing custom scripts to extract telemetry data and unstructured page content from complex, interactive web applications.
 - Configured automated build and deployment pipelines along with static code quality analysis checks, reducing integration issues by 30% and speeding up deployment cycles for microservices.
 - Packaged backend application software using container runtimes and custom installers, securing internal source code and protecting proprietary intellectual property during client deployments.
 - Implemented structured logging across backend services with log rotation policies and severity level controls, simplifying issue debugging and reducing troubleshooting times during outages.

Technical Skills

Languages & Frameworks: Python, Go, Java, SQL, Spring Boot, FastAPI, Gin, REST APIs, gRPC (Spark Connect)

System Design: Microservices, Event-Driven Architecture, Distributed Systems, API Design & Security, Concurrency

Big Data & Databases: Spark, PySpark, Kafka, Airflow, NiFi, Delta Lake, HDFS, PostgreSQL, Solr, Elasticsearch, Redis

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Jenkins, CI/CD, Observability (Prometheus, Grafana, OpenTelemetry), Linux, Bash

Generative AI: LangChain, LangGraph (Agentic Workflows), LLM Integration

Education

- Bachelor of Technology (Electrical & Electronics Engineering) Aug 2018 – Jul 2021
Millennium Institute of Technology & Science, Bhopal 7.59 CGPA

Awards & Achievements

- **Ideathon 2024 Prize** – Secured 2nd Prize (Rs.30,000) in Ideathon 2024 for designing a portable network traffic monitoring system, LightSabre on Laptop.
- **STAR Awards** – Received multiple STAR Awards for contributions to LightSabre, Web Trail, Snapchat, and FlightRadar.